



Precision monitoring of Multiple Sclerosis powered by multi-modal intelligence.



Abhay Naik

Maria Jose
Healey



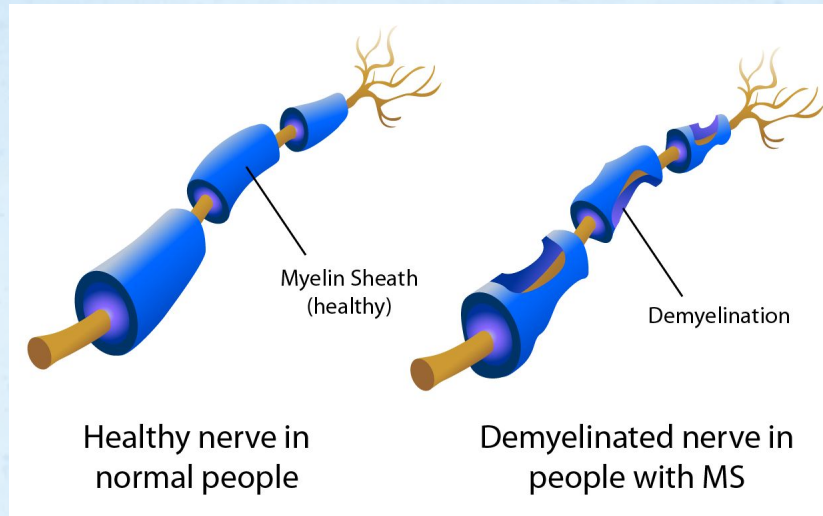
Akshay
Sharathchandra

Surabhi Gupta



Coco Sun

Multiple Sclerosis (MS) is a complex autoimmune disorder



Impacts cognition, sensory systems, and overall independence.

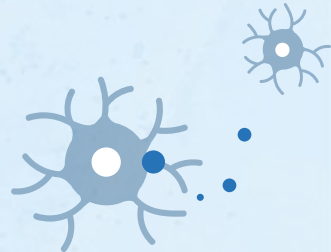
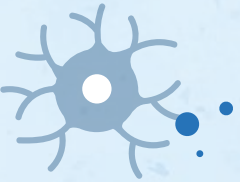
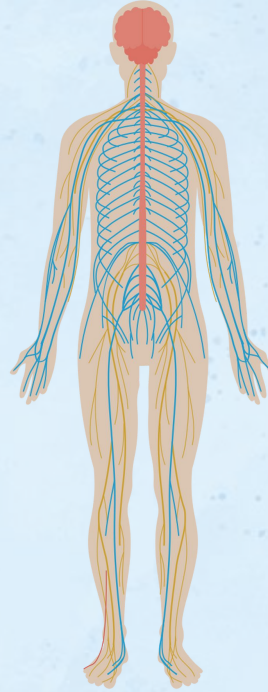
Affects 3 million worldwide, 1 million in the US.

Non-linear, symptoms flare and recede unexpectedly.

MS Data: Valuable but Fragmented

Research

Clinical



MS Data: Valuable but Fragmented

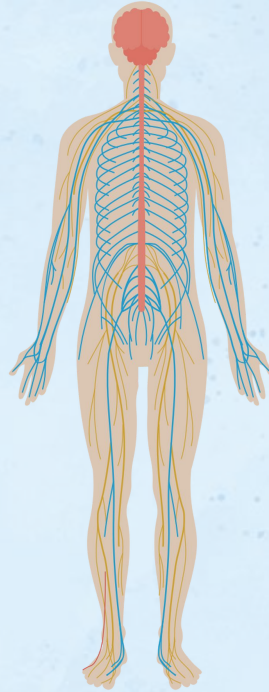
Research



Phage display measures interactions between antibodies and proteins.



NfL measurements quantify the levels of a marker of neuron injury.



Clinical



Clinical attributes are qualitative indicators of progression.

Phage display: measures antibody-protein interactions
NfL: Neurofilament Light Chain

MS Data: Valuable but Fragmented

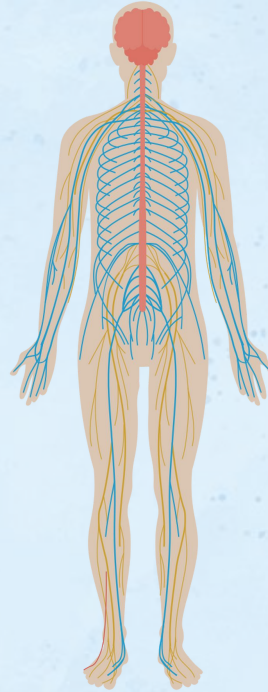
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Clinical attributes are qualitative indicators of progression.

Neurologists are quite effective at treating MS, but we're still trying to figure out *why* it arises.

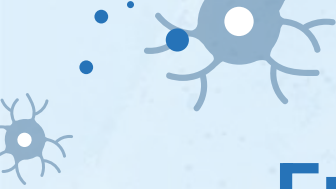
Paraphrased from remarks by Dr. Stephen Hauser, UCSF, 2024.

Phage display: measures antibody-protein interactions
NfL: Neurofilament Light Chain

Our Mission

To empower clinicians and researchers by transforming **proteomic** and clinical data into actionable insights through a reproducible machine learning pipeline, advancing precision care for Multiple Sclerosis patients.

Proteomic data: from proteins



FusionMS: Three Core Innovations



Phage Pipeline

Built a pipeline to analyze phage display data, identifying highly *antigenic* proteins or peptides.



Fusing the Data

Harmonizing **semi quantitative** phage display data, **quantitative** NfL, and **qualitative** clinical attributes.



Predicting DSS

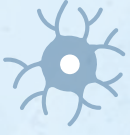
First model combining all metrics to predict disability severity (DSS).

FusionMS: Three Core Innovations



Predicting DSS

First model combining all metrics to predict disability severity (DSS).



FusionMS Demo

See the platform in Action





The Engineering Challenge

"How do we ingest **multi-modal** MS data, run a complex model on it, and return results to a clinician in a way that is **fast, reproducible, and secure?**"



Production Pipeline Architecture

1

Ingest

Node.js Server & S3

2

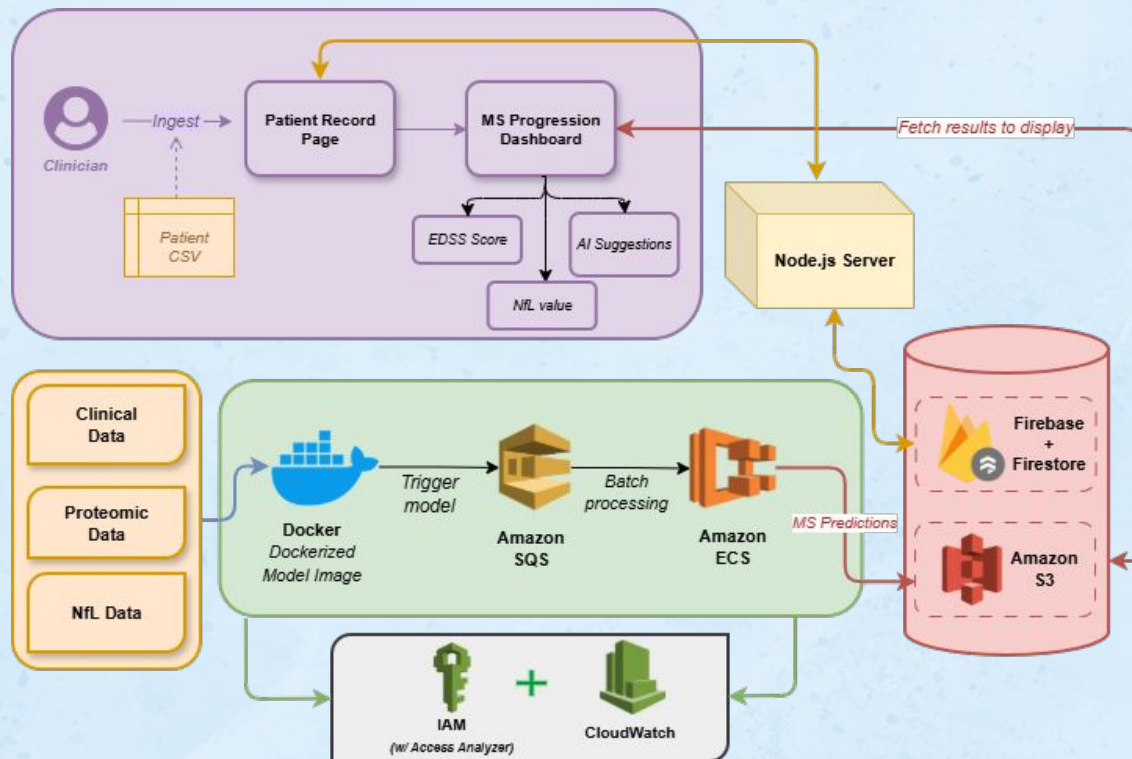
Queue

Amazon SQS Decoupling

3

Compute

ECS (Dockerized Model)



Enterprise-Grade Reliability



IAM Access Control

Strict permissioning for every component. We reason about exactly who can do what, and on which data.



CloudWatch

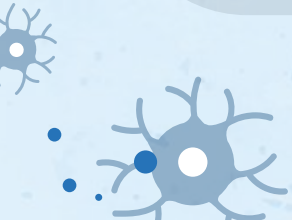
Full observability: logs, metrics, and alerts. Every single run of the pipeline is traceable and monitorable.

The Prediction Questions

Can combining proteomic & clinical data predict patient outcomes?

Key Prediction Questions

1. For a new patient, what will their disability score (DSS) be in 3 years?
2. For a patient at year 3, what will their DSS be in years 4–6?
3. How many patients are fast progressors?



The Data Reality



500 Total Patients



Phage display for ~500k Proteins



NfL Measurements

17

High Value, High Scarcity

166 patients had multi-modal data, but only 17 had sufficient longitudinal records suitable for prediction — highlighting data scarcity challenges common in medical research.

Peptides: fragments of proteins
Phage display: measures antibody-protein interactions
NFL: Neurofilament Light Chain



From Scarcity to Scale

Seed Data

17 Real Patients

Synthetic Data Generation

SDV — HMA Synthesizer
Quality Score: 89.8%

Balance Demographics

SMOTE

1,310
Total Records

SDV — Synthetic Data Vault
HMA — Hierarchical Modeling Algorithm
SMOTE — Synthetic Minority Over-Sampling Technique



1) For a new patient, what will their disability score (DSS) be in 3 years?

Predicting DSS Scores

Explored Elastic Net vs. Hist Gradient Boosting (HGBR), **HGBR** performed best across all metrics.

- Input 1: Top 200 Enriched Protein Signals from Phage Display 
- Input 2: NfL Levels 
- Input 3: Clinical Variables 

Engineered **45** Features: NfL Rate, Age Features, Disease Timing, Symptom Aggregations, Progression Indicators, Interaction Terms

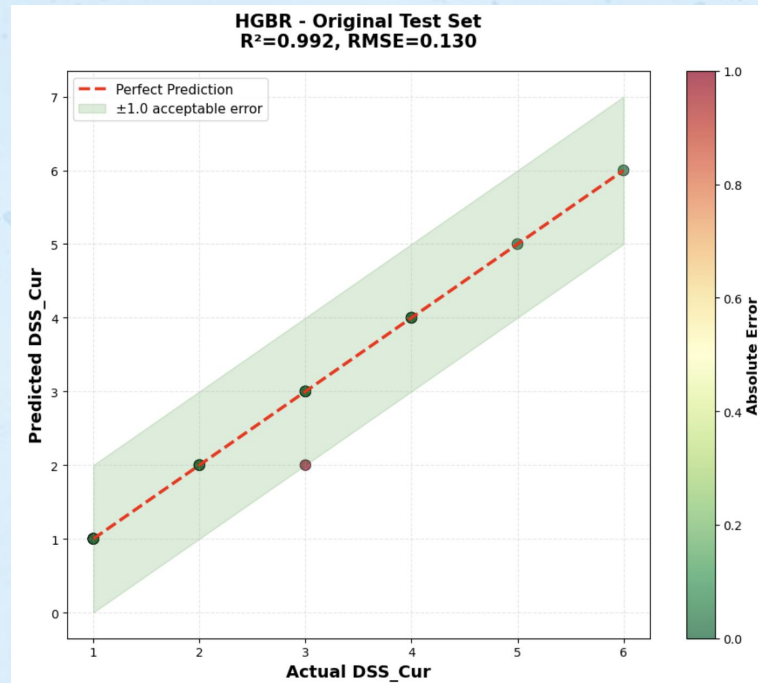
0.05

Mean Absolute Error

100%

Within 1 DSS Point

Peptides: fragments of proteins
Phage display: measures antibody-protein interactions
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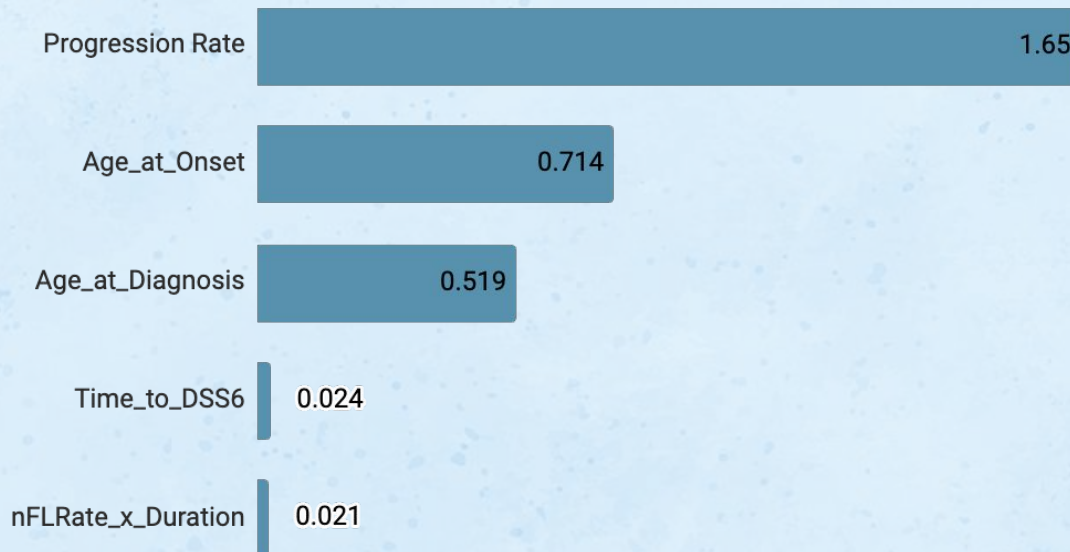


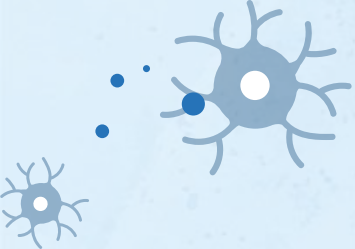


1) For a new patient, what will their disability score (DSS) be in 3 years?

White Box Features Offer Transparency & Interpretability

HBGR Feature Importance





- 2) For a patient at year 3, what will their DSS be in years 4–6?
- 3) How many patients are fast progressors?

Forecasting Disability Progression

Extrapolating DSS scores to anticipate future patient needs.

Extrapolation strategies explored:

- Survival-based model using *Weibull_min* distribution.
- Ensemble approach averaging exponential, logistic, and power progression curves.

No Progression

Predicted DSS Year 3	Extrapolated DSS Year 6	Progression Rate
1	1	0.0

Slow Progression

Predicted DSS Year 3	Extrapolated DSS Year 6	Progression Rate
3	4	0.26

Fast Progression

Predicted DSS Year 3	Extrapolated DSS Year 6	Progression Rate
5	8	1.02

What We Have Achieved



Automated Phage Analysis

Built a robust, end-to-end data processing pipeline.



Multi-Modal Data Ecosystem

Successfully fused complex proteomic (phage display), clinical, and quantitative dataset.



Predicting DSS Outcomes

Developed pioneering ML models for forecasting disability severity (DSS).



Precision Targeting

Delivering patient-level resolution by isolating top 5 enriched peptides.



Comprehensive Dashboard

Built a comprehensive dashboard (**FusionMS**) to surface all integrated insights.

What's Next



Scaling Real-World Data

Integrating live clinical data streams to reduce synthetic dependence and improve model robustness.



GenAI Summaries

Generative AI layer to turn complex predictions into natural-language clinical summaries.



RAG-Enabled Engine

Retrieval-Augmented Generation connecting enriched proteins directly to the scientific publications behind them.



Acknowledgments

Dr. Michael Wilson and team, UCSF Neurology
Dr. Mitchell Wallin, Department of Veterans Affairs

UCSF Weill Institute for
Neurosciences | Department of
Neurology



VA

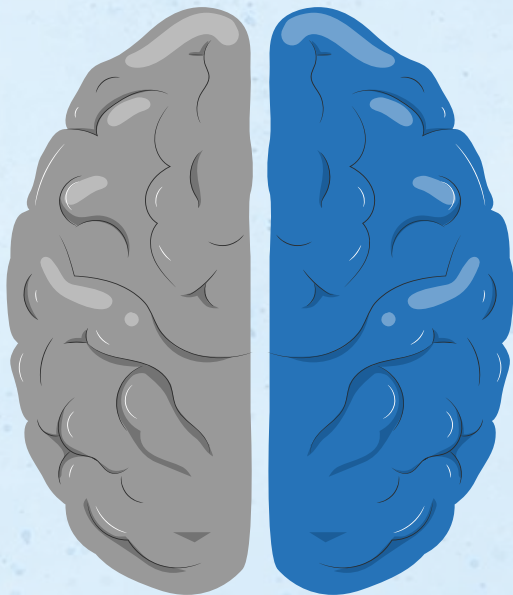
U.S. Department
of Veterans Affairs

The background of the slide is light blue with a subtle pattern of small dots. Several stylized neuron icons are scattered around the text. These icons consist of a central white circle with a dark blue outline, from which several dark blue lines (dendrites and axons) extend outwards. Some neurons are larger than others, and some have small blue dots near their cell bodies.

Fragmented signals become unified intelligence thanks to FusionMS.

We are IntegraMS. And we're just getting started.





Appendix

Feature Engineering: Proteomic data

500k peptides

10x enriched over background
AND Z-score > 3

629-3662 peptides per
sample

Enrichment from pre to post?
(ex: 2x)

Peptide = fragment of protein

Example showing enrichment over background

	patient_1_pre	patient_1_post
Peptide 1	29	78
Peptide 2	2	3
Peptide 3	12	19
Peptide 4	12	68

Feature Engineering: Proteomic data

500k peptides

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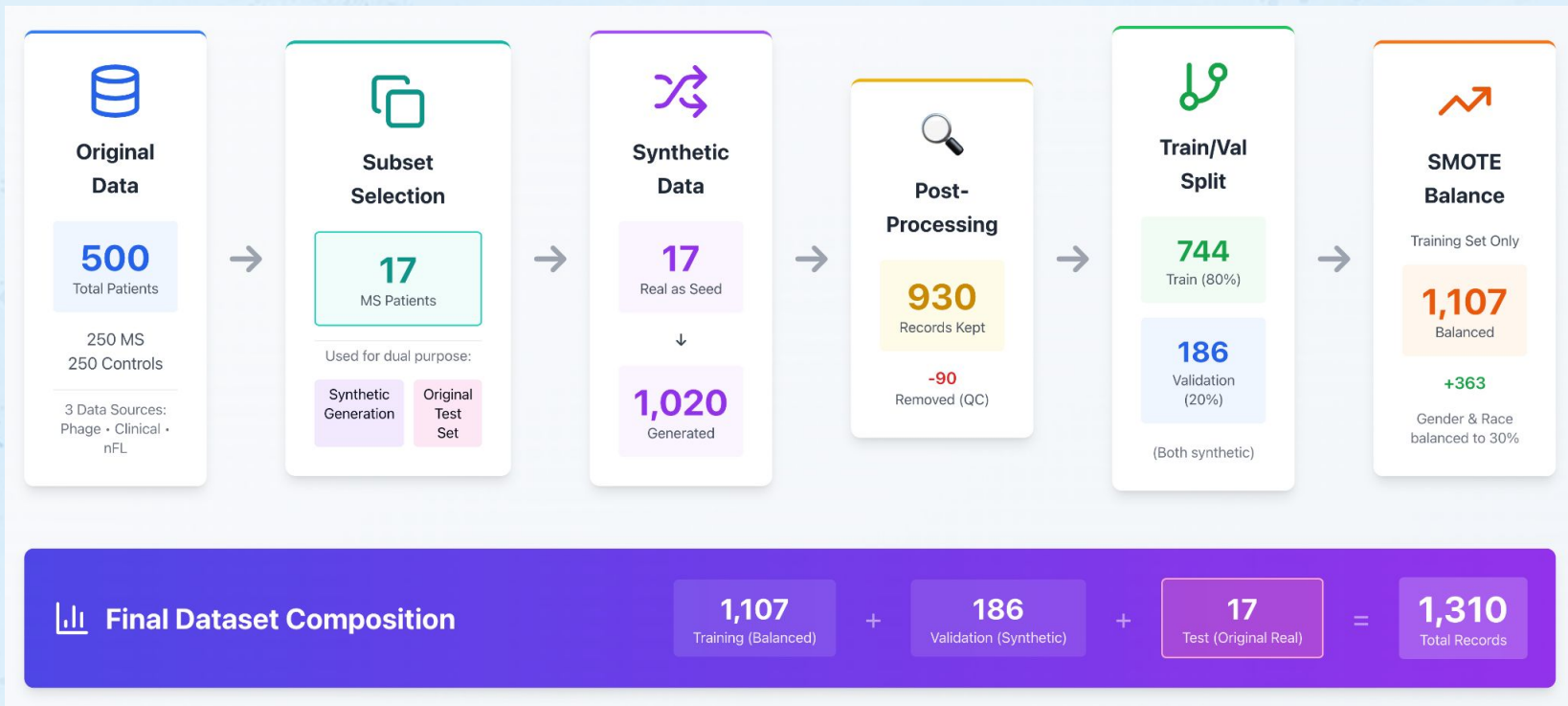
Enrichment from pre to post?
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Peptide = fragment of protein

Example showing enrichment over background

	patient_1_pre	patient_1_post
Peptide 1	29	78
Peptide 2	2	3
Peptide 3	42	49
Peptide 4	12	68

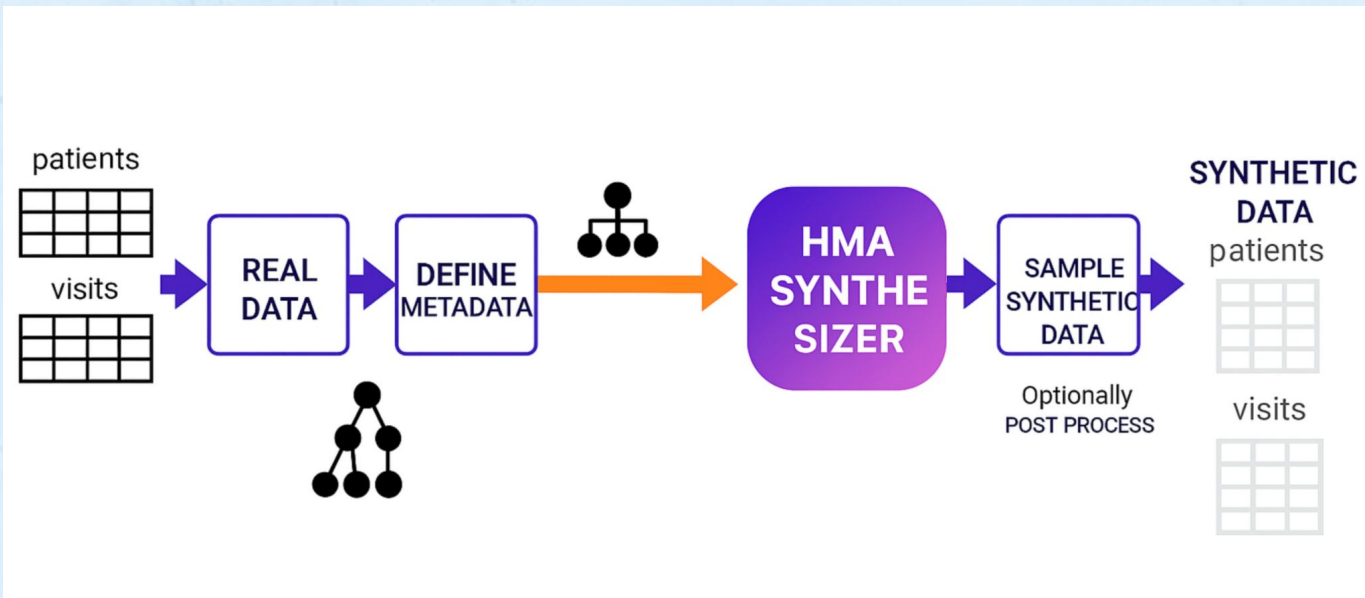
Data Pipeline

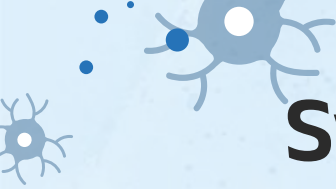


Synthetic Data Generation pipeline

Synthetic Data Vault (SDV)

Multi Table Data - Hierarchical Modeling Algorithm Synthesizer





Synthetic Data Quality Evaluation

Synthetic Data Vault (SDV) - HMA Synthesizer

Synthetic Data Quality Validation

- Overall Quality Score: 89.8% (SDV QualityReport)

Evaluation Metrics:

- Column Shapes: 89.3% — Individual variable distributions match real data
- Column Pair Trends: 84.7% — Correlations and relationships preserved
- Cardinality: 100% — Realistic patient-visit counts maintained
- Intertable Trends: 85.3% — Cross-table patterns replicated

Interpretation: Synthetic patients **closely match** real patients statistically, enabling robust ML model development.



Synthetic Data Post-Processing

Ensuring Biological Plausibility and Data Integrity

1. Temporal Constraints

- $D_Cur \geq D_DX$ (year of current DSS $\geq$ diagnosis year)
- $Ser_Yr > year_birth$ (sample year after birth)
- Visit 2 $Ser_Yr > Visit\ 1\ Ser_Yr$ (chronological ordering)
- $D_onset \leq D_DX$ (disease onset before diagnosis)
- Years post-diagnosis = 3 ($D_Cur = D_DX + 3$)

2. Multi-Table Structure

- 2 visits (pre-onset, post-onset) $\times$ 200 peptides per patient
- Resampled binding signals from real data distribution
- Patient-level attributes consistent across all 400 rows per patient

3. Clinical Constraints

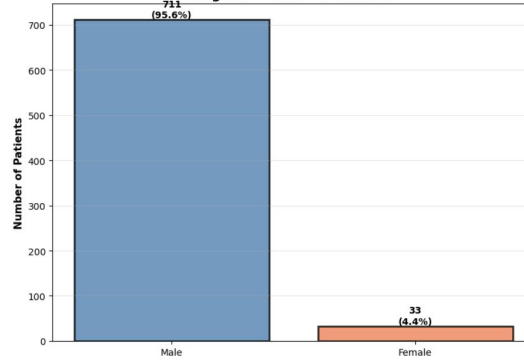
- Race encoding consistency (W $\rightarrow$ 4, B $\rightarrow$ 2)
- DSS milestone logic (if D_DSS6 exists $\rightarrow DSS_Cur=6$, $D_Cur=D_DSS6$)
- Disease timeline features preserved across timepoints

4. Cross-Table Validation

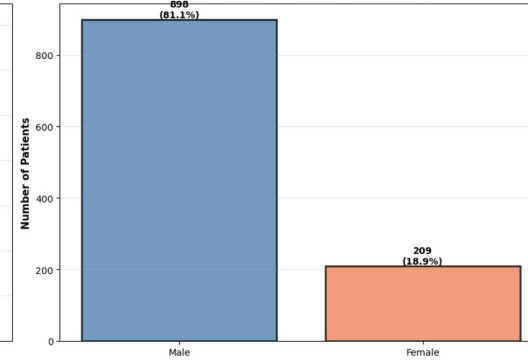
- Patient demographics match across patients and visits tables
- Temporal logic verified for all 1,020 synthetic patients
- 100% consistency achieved post-processing

SMOTE

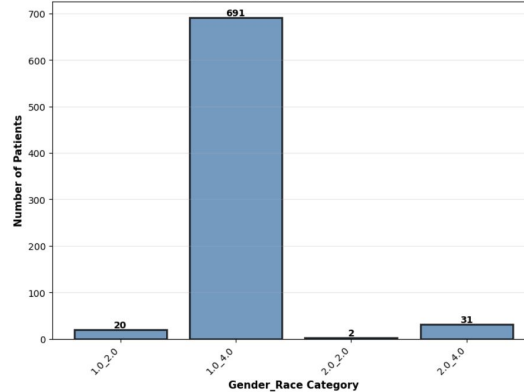
Original Gender Distribution



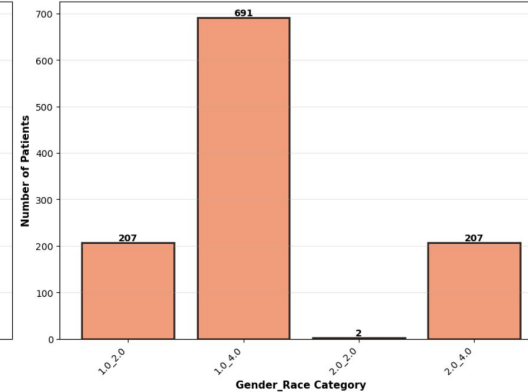
After SMOTE - Gender Distribution



Original Gender-Race Distribution



After SMOTE - Gender-Race Distribution





Forecasting Disability Progression

Extrapolating DSS scores to anticipate future patient needs

Progression Group	Criteria	Number of Patients	Mean DSS Change	Mean Progression Rate
No Progression	0 / year	5	0.00	0.000
Slow	< 0.3 / year	5	0.54	0.166
Moderate	0.3 - 0.7 / year	0	–	–
Fast	>= 0.7 / year	7	2.80	0.859

Patient ID	Predicted DSS Year 3	Extrapolated DSS Year 6	Progression Rate
MS024X	4.893	8.039	0.964
MS079X	4.504	7.227	0.835
MS089X	4.314	6.830	0.771
MS098X	4.114	6.414	0.705
MS106X	4.685	7.606	0.895
MS186X	4.463	7.142	0.821
MS225X	5.070	8.410	1.023

Extrapolations performed using the **Weibull_min** distribution.

The Fragmented Reality



Powerful, yet noisy

Phage display quantifies interactions between autoantibodies and proteins.



Isolated Data

Phage display, clinical attributes, and NfL in blood are measured separately.

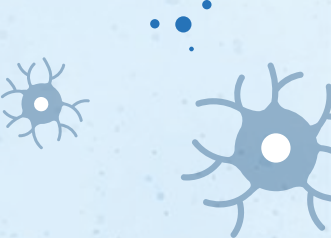


Subjective Analysis

Data analyzed in silos. Qualitative attributes interpreted subjectively.

Neurologists are good at treating MS, but we're still trying to figure out *why* it arises.

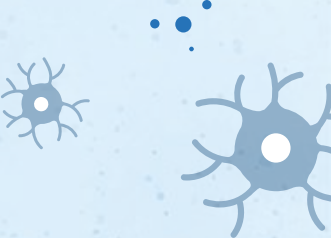
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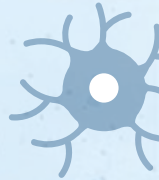
Resources and Documentation

- **MSAA – Understanding Progression in MS** Explains transition from inflammatory relapses to progressive neurodegeneration. mymsaa.org
- **Neurodegeneration in MS: Silent Progression (PMC, 2021)** Shows neurodegeneration occurs early, even during relapsing phases. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8201043/>
- **Neuron (2024): Neurodegeneration and demyelination in MS** Mechanistic evidence that neurodegeneration drives disability accumulation. <https://www.cell.com/neuron/fulltext/S0896-6273%2824%2900372-6?>
- **Giovannoni, G. (2023). Transitioning Between Relapsing and Progressive Multiple Sclerosis: Implications for Treatment Decisions.** Highlights the shift from acute inflammation to chronic neurodegeneration and its impact on treatment strategy. <https://link.springer.com/article/10.1007/s40263-022-00977-3?>
- **SDV documentation**
 - <https://docs.sdv.dev/sdv/multi-table-data/modeling/synthesizers/hmasynthesizer>
 - <https://docs.sdv.dev/sdmetrics/data-metrics/quality/quality-report>
 - <https://docs.sdv.dev/sdmetrics/data-metrics/quality/quality-report/multi-table-api>

Resources and Documentation



- **Synthetic gait data for MS:** Demonstrated how artificially generated gait patterns can augment small clinical datasets, improving the robustness of gait analysis tools for multiple sclerosis patients. <https://arxiv.org/pdf/2411.10377>
- **Synthetic trial replication:** Showed that synthetic datasets can reproduce clinical trial outcomes in MS, offering a cost-effective way to test hypotheses and validate predictive models when patient data is limited. <https://www.researchsquare.com/article/rs-4650837/v1?>
- **Generative AI validation:** Compared synthetic data generated by advanced AI methods against registry data, confirming that synthetic datasets can reliably capture clinical patterns and support exploratory modeling in MS. <https://arxiv.org/pdf/2506.17847>
- **Katz Sand, I. (2021). Classification, diagnosis, and differential diagnosis of multiple sclerosis.** Provides an overview of MS disease course, highlighting the transition from inflammatory activity to progressive neurodegeneration. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8382423/>
- **Reich, D. S., Lucchinetti, C. F., & Calabresi, P. A. (2022). Multiple Sclerosis.** Comprehensive review of MS pathology, progression, and treatment strategies, emphasizing the interplay of inflammation and neurodegeneration. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10186207/>



Resources and Documentation

- **Artificial Intelligence and Multiple Sclerosis** — Review of AI applications in MS, including data challenges and modeling approaches. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10629760/>
- **Predicting Multiple Sclerosis Worsening Using Stratification-Based and Time-Dependent Variables** — Survival modeling of EDSS trajectories to forecast MS progression. https://link.springer.com/chapter/10.1007/978-3-031-82484-5_4
- **SciPy weibull_min Documentation** — Technical reference for the Weibull minimum distribution used in survival analysis and extrapolation. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.weibull_min.html